

Every term of OEIS A397241 is odd

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Abstract

OEIS A397241 is defined by the coefficient condition $n[x^n]A(x)^n = (n-1)[x^n]A(x)^{n+1}$ for $n > 1$, and carries a conjecture of P. D. Hanna that every term is odd. It is true. The condition determines the coefficients one at a time — $a(N)$ enters with coefficient $N^2 - (N^2 - 1) = 1$ — so it determines them modulo 2 as well. Modulo 2 the series $1/(1-x)$, all of whose coefficients are 1, satisfies the condition, because $\binom{2N}{N} = 2\binom{2N-1}{N}$ kills the right side while Kummer's theorem makes $N\binom{2N-1}{N}$ even on the left. Uniqueness then forces $A(x) \equiv 1/(1-x)$, which is the conjecture. The entry's two further conjectures, modulo 3 and modulo 4, are *not* settled here; Section 5 says exactly where the argument stops.

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1 The sequence and the conjecture

OEIS A397241 is “G.f. $A(x)$ satisfies $n * [x^n] A(x)^n = (n-1) * [x^n] A(x)^{n+1}$ for $n > 1$.”. It has offset 0 and begins

1, 1, 1, 11, 191, 5279, 205739, 10703027, ...

The entry gives the expansion

G.f.: $A(x) = 1 + x + x^2 + 11x^3 + 191x^4 + 5279x^5 + 205739x^6 + \dots$

so, writing $A(x) = \sum_{n \geq 0} a(n)x^n$, the entry fixes

$$a(0) = 1, \quad a(1) = 1, \tag{1}$$

and defines the remaining coefficients by

$$n[x^n]A(x)^n = (n-1)[x^n]A(x)^{n+1} \quad \text{for } n > 1. \tag{2}$$

This is the only input taken from the entry.

The entry separately carries the following comment:

Conjecture: all terms are odd. — *Paul D. Hanna*

As of the “Last modified” line on the live entry (Jun 20 2026 05:53:17, revision 12) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way. The entry carries two further conjectures of the same contributor, that $a(n) \equiv 2$ modulo 3 and $a(n) \equiv 3$ modulo 4 for $n > 2$. Those are stronger, they are not proved below, and they are not claimed.

2 The condition determines the sequence

Lemma 1. *Let $A(x) = \sum_{n \geq 0} a(n)x^n$ with $a(0) = 1$. Then for every $M \geq 1$ and every $N \geq 1$,*

$$[x^N]A^M = M a(N) + (a \text{ polynomial with integer coefficients in } a(1), \dots, a(N-1)).$$

Proof. Write $A = 1 + U$ with $U = \sum_{n \geq 1} a(n)x^n$, so $A^M = \sum_{r \geq 0} \binom{M}{r} U^r$. The coefficient of x^N in U^r is a sum of products of r coefficients $a(i_1) \cdots a(i_r)$ with all $i_s \geq 1$ and $i_1 + \cdots + i_r = N$. Such a product involves $a(N)$ only when $r = 1$, and then it is $a(N)$ itself; for $r \geq 2$ every index is at most $N - 1$. Hence $[x^N]A^M = \binom{M}{1}a(N) + \cdots = M a(N) + \cdots$ as claimed. $\square$

Lemma 2. *Given (1), there is exactly one sequence of integers $a(2), a(3), \dots$ satisfying (2), and for each $N \geq 2$ the condition at $n = N$ has the form $a(N) = (a \text{ polynomial with integer coefficients in } a(0), \dots, a(N-1))$.*

Proof. Fix $N \geq 2$ and collect the coefficient of $a(N)$ on the two sides of (2) at $n = N$, using Lemma 1 with $M = N$ and $M = N+1$. The left side gives $N \cdot N$ and the right side $(N-1)(N+1)$, so the net coefficient of $a(N)$ is

$$N^2 - (N^2 - 1) = 1.$$

Everything else in the condition involves only $a(0), \dots, a(N-1)$. So the condition at $n = N$ solves for $a(N)$ over $\mathbb{Z}$, and induction from (1) gives existence and uniqueness. (At $n = 2$ it gives $a(2) = 1$, matching the entry.) $\square$

Corollary 3. *Let $m \geq 2$. If $\bar{A} \in (\mathbb{Z}/m\mathbb{Z})[[x]]$ has $[x^0]\bar{A} = [x^1]\bar{A} = 1$ and satisfies (2) with all coefficients read modulo m , then $a(n) \equiv [x^n]\bar{A} \pmod{m}$ for every n .*

Proof. Reduction modulo m is a ring homomorphism, so it carries the integer recursion of Lemma 2 to the same recursion over $\mathbb{Z}/m\mathbb{Z}$, in which the coefficient of $a(N)$ is still 1 and hence still invertible. That recursion, started from the same two initial values, has a unique solution, and both $a \bmod m$ and the coefficients of $\bar{A}$ solve it. $\square$

3 Two facts about binomial coefficients

Lemma 4. $\binom{2N}{N} = 2\binom{2N-1}{N}$ for every $N \geq 1$.

Proof. Pascal's rule gives $\binom{2N}{N} = \binom{2N-1}{N-1} + \binom{2N-1}{N}$, and $\binom{2N-1}{N-1} = \binom{2N-1}{(2N-1)-(N-1)} = \binom{2N-1}{N}$. $\square$

Lemma 5. *Let $s(N)$ be the number of 1s in the binary expansion of N . Then the exponent of 2 in $\binom{2N}{N}$ is exactly $s(N)$. Consequently $\binom{2N-1}{N}$ is even whenever $s(N) \geq 2$, in particular whenever N is odd and $N \geq 3$.*

Proof. By Kummer's theorem the exponent of 2 in $\binom{2N}{N}$ is the number of carries when N is added to N in base 2; each 1 of N produces exactly one carry, so that number is $s(N)$. By Lemma 4 the exponent of 2 in $\binom{2N-1}{N}$ is $s(N) - 1$, which is at least 1 as soon as $s(N) \geq 2$. Finally, an odd $N \geq 3$ has its last binary digit 1 and at least one further 1, so $s(N) \geq 2$. $\square$

4 The reduction modulo 2

Lemma 6. *Over $\mathbb{Z}/2\mathbb{Z}$ the series $\bar{A}(x) = \frac{1}{1-x}$ satisfies (1) and (2).*

Proof. All coefficients of $1/(1-x)$ equal 1, so (1) holds. For $M \geq 1$, $\bar{A}^M = (1-x)^{-M}$ and $[x^N](1-x)^{-M} = \binom{N+M-1}{N}$, so

$$[x^N]\bar{A}^N = \binom{2N-1}{N}, \quad [x^N]\bar{A}^{N+1} = \binom{2N}{N}.$$

Take $N \geq 2$. By Lemma 4 the right-hand side of (2) is $(N-1)\binom{2N}{N} = 2(N-1)\binom{2N-1}{N} \equiv 0 \pmod{2}$. For the left-hand side $N\binom{2N-1}{N}$: if N is even the factor N is even; if N is odd then $N \geq 3$, so $\binom{2N-1}{N}$ is even by Lemma 5. Either way the left-hand side is $\equiv 0 \pmod{2}$ as well, and (2) holds modulo 2. $\square$

Theorem 7. *Every term of OEIS A397241 is odd. This is the conjecture of Section 1.*

Proof. By Lemma 6 the series $1/(1-x)$ satisfies (1) and (2) over $\mathbb{Z}/2\mathbb{Z}$. By Corollary 3 it is the reduction of A modulo 2, so $a(n) \equiv 1 \pmod{2}$ for every $n \geq 0$. $\square$

5 Where the argument stops

The same scheme would settle the entry's other two conjectures if one could exhibit the corresponding reductions. Those reductions are visible in the data:

$$\bar{A}_3(x) = \frac{1+x^3}{1-x} \text{ over } \mathbb{Z}/3\mathbb{Z}, \quad \bar{A}_4(x) = \frac{1+2x^3}{1-x} \text{ over } \mathbb{Z}/4\mathbb{Z},$$

whose coefficients are $1, 1, 1, 2, 2, 2, \dots$ and $1, 1, 1, 3, 3, 3, \dots$ respectively. Both were checked to satisfy (2) for every $n \leq 34$. What is missing is a proof that they satisfy it for *all* n . Writing $\bar{A}_4 = f + 2h$ with $f = 1/(1-x)$ and $h = x^3/(1-x)$, and using $(2h)^2 \equiv 0$, the condition modulo 4 reduces to

$$(2-N)\binom{2N-1}{N} \equiv 2 \left[N^2 \binom{2N-4}{N-1} - (N^2-1) \binom{2N-3}{N} \right] \pmod{4},$$

and the two sides do agree at every $N \leq 33$ — both vanish unless N or $N-1$ is a power of 2, where both equal 2 — but a proof of that congruence for all N is not given here. The modulo 3 case reduces similarly to a Lucas-type identity. Neither is claimed.

6 Verification

Three checks, each independent of the algebra above.

First, the recursion of Lemma 2 was run in exact rational arithmetic from $a(0) = a(1) = 1$ and compared against the entry's DATA at every one of the 19 published indices; the values agree exactly, and every one is an integer. In particular the condition at $n = 2$ returns $a(2) = 1$.

Second, at every step the coefficient of $a(N)$ was computed rather than assumed, by evaluating the condition at $a(N) = 0$ and at $a(N) = 1$ and subtracting; it came out 1 at every N up to 16, as Lemma 2 requires.

Third, the identity $v_2\left(\binom{2N}{N}\right) = s(N)$ of Lemma 5 was checked directly for every $N \leq 299$, and the reduced condition of Lemma 6 was verified over $\mathbb{Z}/2\mathbb{Z}$ by expanding $(1-x)^{-N}$ and $(1-x)^{-(N+1)}$ for every $n \leq 34$.

References

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